Urban Metabolism of Lima, Peru

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1. Introduction

1.1 Purpose of Report

This document has been prepared with the intention of performing a detailed study of the Urban Metabolism (UM) of Lima, Peru, where the main resource flows, waste outputs, and potential ways of increasing the metabolic efficiency are identified. Furthermore, the report compares two infrastructure interventions: (1) the process of converting organic municipal solid waste (MSW) through anaerobic digestion (AD) to make energy and nutrient-rich digestate, and (2) the conversion of biogenic residuals into low-embodied-carbon construction materials. The report provides information on the quantification of mass and energy flows, the estimation of greenhouse gas (GHG) reductions, and the analysis of the related challenges and strengths of the implementation, proving that urban metabolism can direct engineering interventions in sustainable city planning.

1.2 Unique Regional Quirks and Impact

Lima (12.0467° S, 77.0431° W) is the world's second-largest desert city with a population of about 10 million. It is located on a hyper-arid coastal plain with less than 10 mm of annual rainfall and thus depends mostly on rivers and melting glaciers from the Andes for freshwater. Settlements built in inequitable areas occupy around one-third of the population, and this makes access to water very unequal, leading to some people drawing water from underground sources without permission, and creating complicated waste flows. The fast increase in the number of people living in cities has caused the demand for energy and materials to go up, while the lack of infrastructure and the big informal recycling sector have made the municipal waste management system struggle. Up to 1.8 kg of solid waste is produced per person daily in Lima, and more than half of it is organic. The city's proximity to the ocean and susceptibility to El Niño events exacerbate the problems facing water, power, and waste systems, demonstrating the link between environmental stress, social inequality, and urban metabolism.

1.3 Understanding of "Urban Metabolism"

Urban metabolism regards a metropolis like a living being: it takes in resources (water, energy, food, and materials), changes them through the system (infrastructure and social processes), and throws out waste (wastewater, solid waste, and emissions). Lima's metabolism thus requires quantifying all these flows, which is a major step towards identifying metabolic inefficiencies and possible improvements. Urban metabolism dives deep into the cyclical aspect of using materials and cycling through natural resources. The focus of the report lies on recycling materials through energy recovery and construction material reuse.

2. Visualizing Lima's Urban Region and Its Metabolism

2.1 Regional Context: Climate, Land Use, and Infrastructure

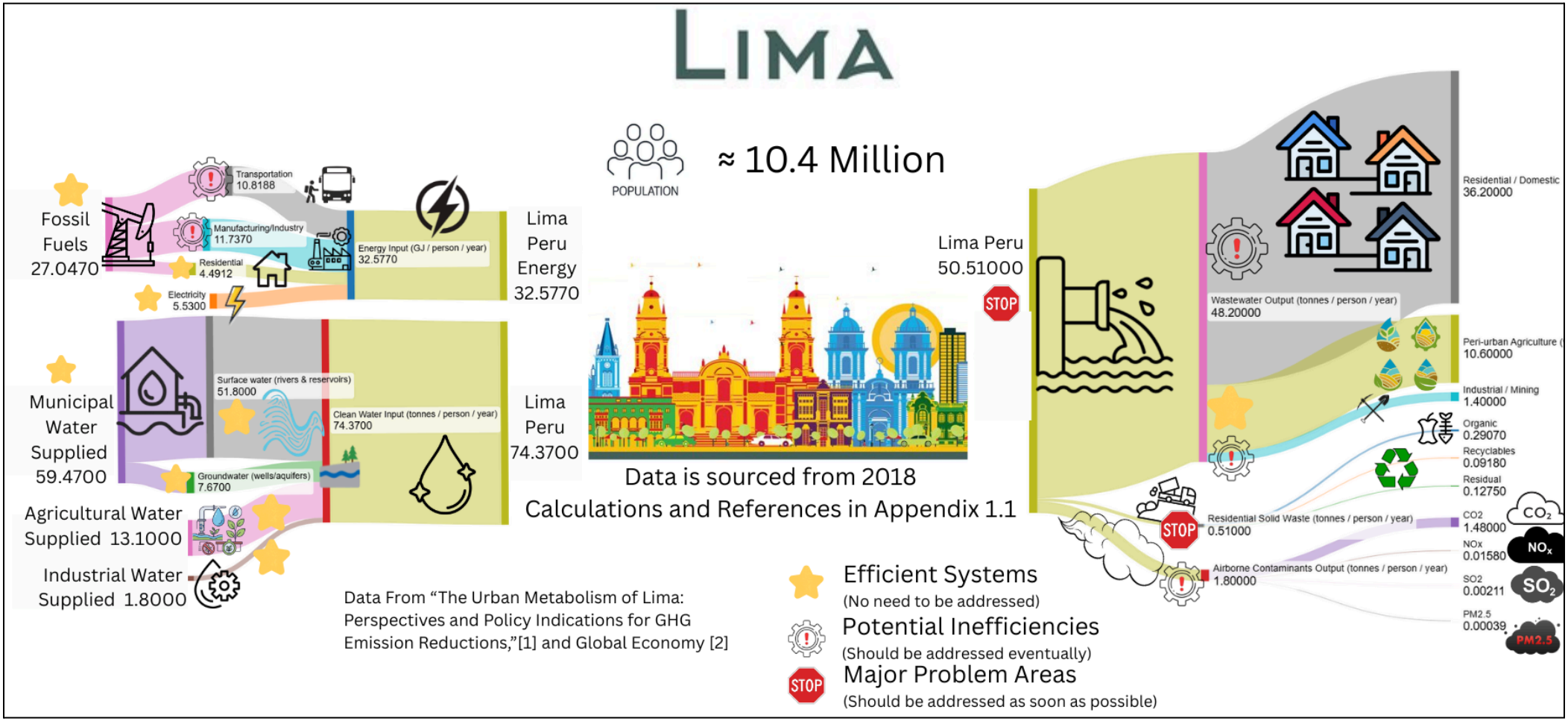

Figure 1. Metabolism diagram of Lima, Peru, outlining inputs and outputs from "The Urban Metabolism of Lima"[1] and Global Economy [2].

An urban metabolism flow diagram in the style of Sankey (Figure 1: Urban metabolism of Metropolitan Lima) visually presents annual flows per capita (GJ/person/yr or tonnes/person/yr) of the main inputs and outputs: Energy input, Clean water input, Wastewater output, Residential solid waste, and Airborne emissions. Numeric labels refer to the best available city-level and metropolitan-level studies (2018); in areas where data were limited, the appendix makes those limitations clear and uses well-documented and conservative estimates.

2.2 Key takeaways/Assumptions highlighted by the diagram:

1. Water: The municipal water supply of Lima is significantly higher in contrast to what the regional renewable supply can provide; the per capita consumption of potable water in the late 2010s is around 59 to 74 tons per person per year (m^3/yr), depending on the report (SEDAPAL/OECD and EW-MFA totals). More than 80% of the supply comes from the Rímac basin, while the other two basins, Chillón and Lurín, provide the rest of the supply.
2. Waste & wastewater: The municipal solid waste generated per person in residential areas is around 0.51 tons per year, and its organic fraction is, according to local composition studies, around 52%. The total of reported wastewater (2014 EW-MFA) is about 42 to 55 tons per person per year, depending on the year and population; the treatment is limited, which leads to a large part of the wastewater being discharged untreated, some of which is reused unofficially in peri-urban agriculture.
3. Energy & emissions: The energy input range is provided by the EW-MFA and the national proxy estimates; the converted EW-MFA total figures imply a very high per-capita primary energy figure (in the order of tens of GJ/person/yr), and CO_2-eq emissions in 2014 were of the magnitude of ~1.48 tCO_2-eq/person/yr (EW-MFA). Transportation and stationary energy are the two main sectors responsible for emissions.

This diagram is integrated across the report: the infrastructure recommendations in Section 3 target the largest, most tractable metabolic flows (organic MSW and its energy/material potential; wastewater reuse and treatment deficits; and transport/energy system carbon intensity).

3. Strategies to Enhance Lima's Urban Metabolism Efficiency

3.1 Common Issues

Municipal solid waste (MSW), the organic / biowaste fraction, is large, widely available, and connects directly to energy, emissions and material flows. Two complementary infrastructure interventions are proposed below.

3.2 Intervention A - City-scale Anaerobic Digestion (AD) for Biowaste → Energy + Digestate

Diverting organics from landfill reduces methane emissions (high global-warming potential), produces usable renewable energy, and yields nutrient-rich digestate that closes nutrient cycles—direct metabolic benefits.

Baseline numbers & assumptions:

- MSW per person ≈ 0.51 t/person/yr (≈230 kg/yr).
- Organic fraction ≈ 52% organic biowaste, ≈ 0.26 t/person/yr.
- Metro population (circa 2018) ≈ 10.39 million, ≈1.24 million t organic/yr

Energy potential (central case):

- Useful energy yield ≈ 2.5 GJ per tonne organic, ≈3.11 million GJ/yr total.
- CHP electricity fraction ≈ 35%, ≈302 GWh/yr electricity equivalent.
- Per-person supply ≈ 29 kWh/person/yr from organics.

GHG reductions (order-of-magnitude):

- Avoided landfill methane (conservative factor 0.25 tCO_2e/t diverted): ~311 $ktCO_2e$/yr avoided.
- Fossil displacement (depending on route) adds ~60–174 $ktCO_2e$/yr avoided (electricity vs thermal displacement), so combined avoided emissions are roughly 0.37–0.49 $MtCO_2e$/yr. Per-person avoided ≈ 0.036–0.047 tCO_2e/p/yr (~36–47 $kgCO_2e$/year).

Implementation (Lima-specific):

- Phase 1 (pilot, years 0-2): The kerbside organics collection will be introduced in a few districts, and a modular anaerobic digestion plant (25-50 kt/yr) will be built at or near the existing transfer station or the market.
- Phase 2 (scale, years 2-6): City scale capacity (in terms of processing the majority of the organics) will be achieved by replicating the modular AD plants, and the integration of CHP into the municipal energy strategy will be followed by upgrading the digestate management.
- Institutions involved: the Municipality and SEDAPAL (for logistics), private operators/PPP for construction and O&M of AD, and universities for monitoring and digestate quality control.

Strengths & weaknesses:

- Strengths: immediate methane reduction; local energy generation; digestate for urban agriculture.
- Weaknesses: needs source separation and contamination control; capital-intensive; digestate markets must be developed; requires sustained policy/behavioural programs.

3.3 Intervention B - Processed Biomass → Low - Embodied - Carbon Construction Materials

Converting stabilized organic residues to durable building products reduces the city's embodied carbon (substituting for fired bricks and cement products), sequesters carbon in materials, and diverts mass from landfills—tightening the material loop in Lima's metabolism.

Baseline & processed mass (conservative):

- From AD digestate wet mass (~0.7 t digestate/t input) to 1.24 Mt input, ≈0.87 Mt wet digestate/yr. After drying and stabilization, ~250–300 kt/yr dry solids could be available for industrial use (order-of-magnitude, process dependent).
- Target substitution: use ~150 kt/yr of dried digestate as feedstock for compressed bio-blocks and stabilized soil composites (pilot-to-medium program).

Quantified benefits:

- Assume avoided embodied carbon ≈ 0.2 tCO_2e avoided per tonne of conventional material replaced by 150 kt × 0.2 = 30 $ktCO_2e$/yr avoided.

- Co-benefit: sequestration of some biogenic carbon in durable products (subject to product lifespan and end-of-life treatment).

Lima Specific Implementation:

- Small-scale production facilities near AD drying plants; R&D/testing partnerships with local universities (material strength, leaching, durability); municipal procurement targets for public works to create demand; pilot projects for non-structural elements (paving, partition walls) moving to larger structural applications after certification.

Strengths & weaknesses:

- Strengths: reduces landfill mass; creates jobs and local industry; substitutes high-carbon inputs.
- Weaknesses: requires rigorous quality control and standards; feedstock variability; market acceptance risk; processing/drying energy needs (can be supplied with low-grade heat from AD to reduce fossil needs).

Combined systemic impact (interactions & scale)

- AD + biomass substitution are synergistic: AD provides energy and a stable feedstock for materials. Combined, these interventions divert most organic MSW, produce energy (~300 GWh/yr), reduce landfill CH_4, and avoid embodied carbon in construction (~30 ktCO_2e/yr targeted substitution). The interventions target hard metabolic leverage points: large organic mass, close connection to energy and materials, and existing peri-urban demand for compost/substrate and construction feedstock.

Quantitative summary (central estimates):

- Organic mass processed: ~1.24 Mt/yr.
- Electricity from organics: ~302 GWh/yr.
- Avoided GHGs: ~0.37–0.49 MtCO_2e/yr (AD + fossil displacement + landfill avoidance) + ~0.03 MtCO_2e/yr (material substitution) ≈ 0.40–0.52 MtCO_2e/yr total city-wide avoided emissions (range & sensitivity in appendix).

4. Critique of the Urban Metabolism Analogy

4.1 Validity & usefulness of the analogy

The urban metabolism framework acts as an industrial engineer's best friend, as it leads to the understanding of stocks and fluxes of the urban metabolism, the aggregation of flows, pinpointing of the major ones (like organic mass, water supply), and the clear indications of the infrastructure changes that would bring the desired outcome. In the case of Lima, UM reveals the following: (a) the organic fraction of MSW is a large, concentrated flow; (b) the water supply is dependent on the basin and is at risk of being cut; and (c) lack of proper treatment of wastewater results in poor water quality and health hazards. The AQ + biomass building projects are thus directly driven by these revelations.

Where the analogy works well.

- Material and energy accounting: The UM is the best in quantifying inputs and outputs not only of the water, waste and energy systems but also of the physical infrastructure capacity and the failure modes (treatment plants, landfills, water supply), among others.
- Policy prioritization: The UM points out high-mass, high-impact flows (for example, organics) where the change in the infrastructure results in significant system-level advantages.

Where the analogy falls short / caveats.

- Spatial & social heterogeneity: UM tends to aggregate. This can hide inequalities and local capacity constraints (informal settlements, differential access to services). For Lima, peri-urban reuse practices or informal waste pickers are critical actors that aggregate metrics can obscure.
- Temporal dynamics & behaviour: UM is portrayed as a steady-state, not capturing short-term peaks (seasonal water scarcity) or behaviour change dynamics required for source separation.
- Ecological vs social functions: cities are socio-technical systems. Having a city treated like an organism risks underplaying governance, cultural, and institutional determinants of flows.

Assessment verdict (engineering perspective).

Despite limitations, UM still gives great importance to engineering planning in Lima: the system gives importance to changes of large flows and later assists in estimating the overall benefits that would result from infrastructure modifications (shown in Section 3). At the same time, the limitations of UM can still be overcome by pairing up the accounting of UM with engagement of local stakeholders, GIS spatial analysis, and pilot testing for revealing heterogeneity and constraints of behaviour.

5. Recommendations and Conclusions

1. Anaerobic Digestion: Implement staged organics diversion to feed modular AD facilities (~1.2 Mt/yr). Benefits: 300 GWh/yr electricity potential, 0.37-0.48 $MtCO_2e$/yr avoided.
2. Biomass Construction Materials: Process dried digestate into compressed bio-blocks (~150 kt/yr). Benefits: ~30 $ktCO_2e$/yr avoided, local jobs, material circularity.
3. Supportive Policies: Mandatory source separation, municipal procurement mandates, landfill diversion targets, and public education campaigns. Adopt a staged organics diversion + AD program (pilot in markets/districts to scale to the city). Prioritize source separation programs and modular AD plants co-located with transfer stations.
4. Fill data gaps: commission a city-wide sectoral wastewater and emissions inventory (2018 baseline update) and mapping of district-level MSW composition to refine facility sizing and economic modelling. UM analysis should be iteratively updated with measured pilot results.

Lima's UM analysis demonstrates that targeted infrastructure interventions can significantly reduce waste, generate renewable energy, and create a circular economy for organic residues, supporting both climate and social co-benefits.

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Appendix

1.1 Input/ Output with Calculations (Chosen Year 2018

Flow (units)	2014 (value source year)	2018 (value source year)	2020 (value source year)	Notes/source & calculation (one-line)
Population (Metropolitan Lima)	**9.63 million** (2014). (macrotrends.net)	**10.39 million** (2018). (macrotrends.net)	**10.72 million** (2020). (macrotrends.net)	MacroTrends metro Lima population series used for per-capita conversions.
ENERGY INPUT (GJ/person/year)				
• **Total primary energy use (proxy: national kgoe → GJ)**	**32.24 GJ / person / yr** (2014 kgoe = **767.69** → 0.76769 toe ×42 = 32.24 GJ). (CEIC Data)	**≈32.6 GJ/person/yr** (2018 national per-capita energy series, converted). (TheGlobalEconomy.com)	**≈34.7 GJ/person/yr** (2020 national per-capita energy series, converted). (TheGlobalEconomy.com)	**Method/caveat:** City-level GJ/person rarely published → used national primary energy use (World Bank / CEIC / OurWorldInData) as a conservative proxy and converted (1000 kgoe = 42 GJ). (CEIC Data)
• Fossil fuels (split) **(GJ/person/yr)**	**27.533 GJ/person/yr (2014)**	**27.047 GJ/person/yr (2018)**	**29.555 GJ/person/yr (2020)**	National statistics show fossil share, but not a city-level breakdown in the cited sources. (CEIC Data)
• Electricity (stationary) **(GJ/person/yr)**	**~4.30 GJ/person/yr (2014)** (per earlier estimate using reported electricity kTOE equivalent 986 ktoe → convert). (Frontiers)	**~5.53 GJ/person/yr (2018)**	**~5.145 GJ/person/yr (2020)**	Electricity value derived from city energy reports cited previously (Facchini / Table values for electricity). (Frontiers)
CLEAN WATER INPUT —				

(tonnes/person/year)				
• Total municipal water supplied	**≈55.7 t / person/yr** (2014 / **536 million m³** total water consumption ÷ 9.63M = 55.7 m³/person = 55.7 t). (Frontiers)	**≈59.5 t / person/yr** (2018 OECD/SEDAPAL estimates ~163 L/day → 0.163×365 = 59.5 m³/yr). (EU Trade)	**≈60 t / person/yr** (2020 SEDAPAL/World Bank reporting & trend / reported totals ~same order). (World Bank)	**Method/caveat:** 1 m³ = 1 tonne (water). 2014 total water from Facchini (Frontiers). 2018 uses OECD/SEDAPAL per-day numbers. (Frontiers)
• Agricultural water supplied	**N/a**	**≈13.1 t / person/yr** (EU Trade)	**N/a**	
• Industrial water supplied	**N/a**	**≈1.8 t / person/yr** (EU Trade)	**N/a**	
WASTEWATER OUTPUT (tonnes/person/year)				
• Municipal wastewater (collected/removed)	**≈42.5 t / person/yr** (2014, Table values vary; Wiley Table 2 lists wastewater **409,724 kt** for 2014 → 409,724,000 t ÷ 9.63M = 42.55 t). (Frontiers)	**≈48.1-53.1 t /person/yrr** (2018, using reported sewer flows / larger pop ~10.39M; sewer collection figures from SEDAPAL / IWMI baseline ~17–18 m³/s → annual ≈551–567M m³ → ÷ pop). (sdcrrr.iwmi.org)	**≈51-55 t / person/yr** (2020, ballpark from city sewer volumes & pop). (sdcrrr.iwmi.org)	**Method/caveat:** Some sources report *supplied water* (Facchini), and other sources report *wastewater collected* (Wiley Table / SEDAPAL / IWMI). Not all supplied water is collected (leakage / informal use). I used reported wastewater totals where available. (Frontiers)
RESIDENTIAL SOLID WASTE (tonnes/person/year)				

• Total residential/municipal solid waste (per-capita)	**≈0.51 t / person/yr** (2014-2018 mid estimate; 0.57–0.77 kg/person/day in referenced sources → central 0.63 kg/d → 0.63×365 = 0.230 kg = 0.230 t/yr). (Holland Circular Hotspot)	**≈0.51–0.53 t / person/yr** (2018 national MSW 0.63 kg/d used; Lima typically similar or a little higher). (Holland Circular Hotspot)	**≈0.51–0.56 t / person/yr** (2020 national reports 0.57–0.77 kg/d range; converted). (Holland Circular Hotspot)	**Method/caveat:** Converted kg/person/day → t/person/yr. Official collection/recording may underreport informal disposal; Lima's GPC (generation per capita) varies by district. (Holland Circular Hotspot)
• Breakdown applies national/Lima composition to the total (use 2018 Peru composition ≈ 57.7% organic / 18.3% recyclables / 24.1% residual)	Organic ≈0.29 t/p/yr Recyclables ≈0.091 t/p/yr Residual ≈0.12 t/p/yr	**(same composition applied)** Organic ≈0.29-0.30 t; Recyclables ≈0.091–0.1 t; Residual ≈0.12–.013 t	**(same)** Organic ≈0.29-0.31 t; Recyclables ≈0.091–0.11 t; Residual ≈0.12–0.14 t	Source for composition: national MSW composition reports (Holland/Minam/SIGERSOL datasets), Peru ~57.7% organic, 18.3% recyclables, 24.1% residual (2018). Use these proportions unless district data is available. (Holland Circular Hotspot)
AIRBORNE CONTAMINANTS OUTPUT (tonnes/person/year)				
• Total GHG emissions (CO_2-eq)	**≈1.48 t CO_2-eq / person / yr** (2014 / Table 2 / EW-MFA lists **14,229 ktCO_2-eq** for 2014 → 14,229,000 t ÷ 9.63M = 1.48 tCO_2-eq/p/yr). (Frontiers)	**≈1.48 t CO_2-eq / person / yr (2018)**	**use 2018**	**Method/caveat:** CO_2-eq given in Wiley Table 2 (EW-MFA) for 2014; earlier vehicular inventories give species (NOx, SO_2, PM) for 2018, see below. (Frontiers)

• NO_x (vehicle inventory)	**(not reported in 2014 EW-MFA)**	**≈15.8 kg NO_x / person/yr** (2018 / vehicle inventory total **164,038 t NO_x ÷ pop 10.39M = 15.79 kg/p/yr**). (Environmental Protection Agency)	**(Use 2018 inventory as a proxy for 2020 unless a later inventory is available.)**	Source: EPA / Lima Megacities vehicle emissions inventory (2018) vehicle totals for PM2.5/NOx/SO2. (Environmental Protection Agency)
• SO_2 (vehicle inventory)	**use 2018**	**≈2.11 kg SO_2 / person / yr** (2018 / **21,908 t SO_2 ÷ 10.39M = 2.11 kg/p/yr**). (Environmental Protection Agency)	**use 2018**	This is *vehicle* SO_2 only; industrial/port/shipping sources add more. (Environmental Protection Agency)
• $PM_{2.5}$ (vehicle inventory)	**use 2018**	**≈0.39 kg $PM_{2.5}$ / person / yr** (2018 /**4,092 t ÷ 10.39M = 0.394 kg/p/yr**). (Environmental Protection Agency)	**use 2018**	Vehicle inventory $PM_{2.5}$ only; ambient PM includes other sources (industry, dust, wood burning). (Environmental Protection Agency)

1.2 Table: Air Pollutants

Sector	Pollutant / GHG	Estimated contribution per person (~2018)	Source / Notes
Transport	CO_2-eq	**~0.40 × (total per-person)**	Using a 40% share from Lima Province's GHG inventory. If total emissions ~15.79 Mt CO_2-eq / ~10 M people → ~1.58 t CO_2-eq/person; 40% → ~0.63 t/person.
Stationary energy (industrial + buildings)	CO_2-eq	**~0.40 × (total per-person)**	Same source as above. Using same calculation logic → ~0.63 t/person (if using the same total)
Waste	CO_2-eq	**~0.20 × (total per-person)**	From Lima inventory. Using the same base → ~0.32 t/person.

Transportation	**$PM_{2.5}$**	**~0.000394 t/person/year** (≈ 0.394 kg/person/yr)	From 4,092 tons $PM_{2.5}$ (2018) in vehicle emissions divided by ~10.39 M people. This is from transport only.
Transportation	**NO_x**	**~0.0158 t/person/yr** (≈ 15.8 kg)	164,038 t NO_x ÷ ~10.39M people.
Transportation	**SO_2**	**~0.00211 t/person/yr** (≈ 2.11 kg)	21,908 t SO_2 ÷ ~10.39M.

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